

Cyberphysical Substrate: Canvas, Body, Archive

Cyberphysical Substrate: Canvas, Body, Archive I

Cyberphysical DigiPPPiP treats the “paper” in PPPiP as a variable substrate rather than a fixed object. Pen-and-paper remains the haptic baseline, but shared tablets, web canvases, augmented reality (AR) overlays, virtual reality (VR) drawing spaces, smart paper, and persistent whiteboards all create different couplings among gesture, mark, archive, and partner response. Cyber-physical-social systems (CPSS) research is useful because it keeps physical, cyber, and social layers in one analytic frame rather than reducing the system to a device or app [sobb2023cpss]. Collaborative-work studies add a second discipline: the drawing surface must preserve the interactional value of gestures, spatial relations, and work-in-progress marks, not only the final image [tang1991collaborativework; ishii1993clearboard].

Cyberphysical Substrate: Canvas, Body, Archive II

That design discipline is stronger when the canvas is treated as a shared-workspace system rather than as a drawing file with two accounts. Workspace awareness asks what each partner can know about the other's location, action, intention, and change history in the shared field [gutwin2002workspaceawareness]. Social translucence asks whether that visibility supports social coordination and accountability without becoming total visibility [erickson2000socialtranslucence]. Collaborative tabletop research adds a spatial constraint: people create personal, group, and storage territories even when the work surface is shared, so the canvas should support mutual access without erasing ownership, reach, and turn structure [scott2004territoriality]. DigiPPPiP should therefore expose partner activity at the granularity needed for response: live strokes, recent changes, viewport differences, consent state, and replay availability. It should not expose hidden drafts, every hesitation, device telemetry, or physiological streams by default.

Cyberphysical Substrate: Canvas, Body, Archive III

Digital drawing should therefore be evaluated by affordance fit rather than by whether it perfectly imitates paper.

Context-sensitive pen-tablet systems show that drawing can become more natural when software is aware of the relation among hand, pen, and surface [sun2011pentablet]. Remote co-design work likewise shows that shared digital sketching can support collaborative understanding even when the expressive qualities of paper remain distinctive [close2024codesign]. Creativity-support and collaborative-visual-analytics scholarship adds a methods lesson: tools should support exploration, annotation, provenance, shared context, and recoverable interpretation rather than only faster production [shneiderman2007creativity; heer2008collaborative]. Co-creative drawing workflow research adds an important boundary for artificial intelligence (AI) enhancements: computational systems can observe, model, and prompt creative processes, but they should be evaluated by how they support human authorship and shared interpretation [jansen2021cocreative]. DigiPPPiP extends these design lessons into intimate and domestic contexts where low friction, privacy, ambiguity, and ritual